

Group & Ring Theory MATP33: Course Summary of Important Results

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Lecture 0 — Background Results

Theorem 1 (First Isomorphism Theorem). *Let $f : G \rightarrow H$ be a group homomorphism. Then $\ker f \trianglelefteq G$ and $\operatorname{im} f \cong G/\ker f$.*

Theorem 2 (Second Isomorphism Theorem). *Let G be a group, $H \leq G$, and $N \trianglelefteq G$. Then $HN \leq G$, $H \cap N \trianglelefteq H$, and $HN/N \cong H/(H \cap N)$.*

Theorem 3 (Third Isomorphism Theorem). *If $K \trianglelefteq G$ and $K \leq L \trianglelefteq G$, then $(G/K)/(L/K) \cong G/L$.*

Theorem 4 (Correspondence Theorem). *Let $N \trianglelefteq G$. Every subgroup of G/N is of the form H/N where $N \leq H \leq G$, and conversely if $N \leq H \leq G$ then $H/N \leq G/N$.*

Theorem 5 (Lagrange). *If $H \leq G$ and G is finite, then $|H|$ divides $|G|$.*

Theorem 6 (Fundamental Theorem of Finitely Generated Abelian Groups). *Let A be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

where $s \geq 0$ and $a_1 \mid a_2 \mid \cdots \mid a_r$ are non-zero non-units in $\mathbb{Z}$. This decomposition is unique.

Lecture 1

Theorem 7 (1.10). *Let G be a group and X a set. Then:*

- (i) If X is a G -set, then the action of G on X induces a homomorphism $\varphi : G \rightarrow S_X$;*
- (ii) Any homomorphism $\varphi : G \rightarrow S_X$ induces an action of G on X .*

Proof. (i) Define $\varphi : G \rightarrow S_X$ by $(\varphi(a))(x) = a \cdot x$ for $a \in G$ and $x \in X$. We must first check that $\varphi(a) \in S_X$ for each $a \in G$, i.e. that the function $x \mapsto a \cdot x$ is a bijection $X \rightarrow X$.

- *Injective:* Suppose $a \cdot x = a \cdot y$. Then $x = a^{-1} \cdot (a \cdot x) = a^{-1} \cdot (a \cdot y) = y$.
- *Surjective:* For any $x \in X$, we have $a \cdot (a^{-1} \cdot x) = x$, so $x \in \operatorname{im}(\varphi(a))$.

Hence $\varphi(a) \in S_X$. To see φ is a homomorphism, let $a, b \in G$. Then for all $x \in X$,

$$(\varphi(ab))(x) = (ab) \cdot x = a \cdot (b \cdot x) = (\varphi(a))((\varphi(b))(x)) = (\varphi(a)\varphi(b))(x).$$

(ii) Define $a \cdot x := (\varphi(a))(x)$. Then $(ab) \cdot x = (\varphi(ab))(x) = (\varphi(a)\varphi(b))(x) = a \cdot (b \cdot x)$, and $e \cdot x = (\varphi(e))(x) = x$. Hence X is a G -set. $\square$

Theorem 8 (1.11 — Cayley's Theorem). *Every group G is isomorphic to a subgroup of the symmetric group S_G .*

Proof. By Theorem 1.10(i), left translation $a \cdot x := ax$ gives a homomorphism $\varphi : G \rightarrow S_G$. If $a \in \ker \varphi$, then $\varphi(a) = \text{id}_G$, so $ax = x$ for all $x \in G$, giving $a = e$. Hence $\ker \varphi = \{e\}$, and by the First Isomorphism Theorem $G \cong G/\ker \varphi \cong \text{im } \varphi \leq S_G$. $\square$

Definition 1 (Normal core). *Let G be a group and $H \leq G$. The normal core of H in G is*

$$\text{core}(H) := \bigcap_{x \in G} xHx^{-1}.$$

It is the largest normal subgroup of G contained in H .

Theorem 9 (1.12). *Let G be a group and $H \leq G$ a subgroup of index n . Then there is a homomorphism $\varphi : G \rightarrow S_n$ such that $\ker \varphi = \text{core}(H)$.*

Proof. Since $|G/H| = n$, we have $S_{G/H} \cong S_n$. By Theorem 1.10(i), the action of G on the coset space G/H by left multiplication gives a homomorphism $\varphi : G \rightarrow S_n$ with $\varphi(a)(xH) = axH$. Then

$$\begin{aligned} \ker \varphi &= \{a \in G \mid axH = xH \text{ for all } x \in G\} \\ &= \{a \in G \mid x^{-1}ax \in H \text{ for all } x \in G\} \\ &= \{a \in G \mid a \in xHx^{-1} \text{ for all } x \in G\} \\ &= \bigcap_{x \in G} xHx^{-1} = \text{core}(H). \end{aligned} \quad \square$$

Corollary 1 (1.13). *Let G be a group with a normal subgroup H of index n . Then G/H is isomorphic to a subgroup of S_n .*

Proof. Since $H \trianglelefteq G$, conjugation fixes H , so $\text{core}(H) = \bigcap_{x \in G} xHx^{-1} = H$. By Theorem 1.12, $\ker \varphi = H$, and by the First Isomorphism Theorem $G/H \cong \text{im } \varphi \leq S_n$. $\square$

Theorem 10 (1.24 — Orbit-Stabiliser Theorem). *Let G be a group acting on a set X . Then the set of all orbits in X under G is a partition of X . Furthermore, for each $x \in X$ there is a bijection $Gx \rightarrow G/G_x$, and hence*

$$|Gx| = |G : G_x|.$$

Therefore, if X is finite,

$$|X| = \sum_{x \in C} |G : G_x|,$$

where $C \subseteq X$ contains exactly one element from each orbit.

Proof. (Partition.) Define $x \sim y$ if $x = ay$ for some $a \in G$. This is an equivalence relation: reflexivity follows from $e \cdot x = x$, symmetry from $x = ay \Rightarrow y = a^{-1}x$, and transitivity from $x = ay, y = bz \Rightarrow x = (ab)z$. The equivalence class of x is exactly the orbit Gx , so the orbits partition X .

(*Bijection* $Gx \rightarrow G/G_x$.) Define

$$\varphi : Gx \rightarrow G/G_x, \quad ax \mapsto aG_x.$$

The key observation is the following chain of equivalences, for any $a, b \in G$:

$$ax = bx \iff a^{-1}bx = x \iff a^{-1}b \in G_x \iff aG_x = bG_x.$$

- *Well-defined:* If $ax = bx$ (the same orbit element written using two different group elements), then $aG_x = bG_x$, so the coset φ assigns does not depend on the choice of representative.
- *Injective:* If $\varphi(ax) = \varphi(bx)$, i.e. $aG_x = bG_x$, then $ax = bx$ by the chain above.
- *Surjective:* Take any coset $aG_x \in G/G_x$. Since $a \in G$, we have $ax \in Gx$, and $\varphi(ax) = aG_x$.

Hence φ is a bijection, giving $|Gx| = |G/G_x| = |G : G_x|$. Summing over one representative per orbit gives $|X| = \sum_{x \in C} |G : G_x|$. $\square$

Theorem 11 (1.25 — Class Equation). *Let G be a finite group. Then the conjugacy classes partition G , and*

$$|G| = \sum_{a \in C} |G : C_G(a)|,$$

where $C \subseteq G$ contains exactly one element from each conjugacy class.

Proof. Let G act on itself by conjugation, i.e. $a \cdot x = axa^{-1}$. The orbit of x under this action is its conjugacy class $\text{ccl}(x) = \{axa^{-1} \mid a \in G\}$, and the stabiliser of x is $G_x = \{a \in G \mid axa^{-1} = x\} = C_G(x)$. Applying Theorem 1.24 directly gives the partition into orbits and $|G| = \sum_{a \in C} |G : C_G(a)|$. $\square$

Lecture 2

Theorem 12 (2.1). *Let G be a finite group of order p^n , where p is prime and $n \in \mathbb{N}$. Then:*

- (i) G has a non-trivial centre $Z(G)$;
- (ii) $Z(G) \cap K$ is non-trivial for any non-trivial normal subgroup K of G ;
- (iii) if H is a proper subgroup of G , then $H \subsetneq N_G(H)$. In particular, if $|H| = p^{n-1}$ then $H \trianglelefteq G$.

Proof. (i) Since $x \in Z(G)$ iff $\text{ccl}(x) = \{x\}$, the class equation refines to

$$p^n = |Z(G)| + \sum_{x \in C} |G : C_G(x)|,$$

where C runs over conjugacy classes with more than one element. For each $x \in C$, $|C_G(x)|$ properly divides p^n , so $p \nmid |G : C_G(x)|$. Hence p divides the sum, and therefore $p \mid |Z(G)|$. Since $e \in Z(G)$, we have $|Z(G)| \geq 1$, so in fact $|Z(G)| \geq p > 1$.

(ii) Intersecting the class decomposition of G with K gives $|K| = |Z(G) \cap K| + \sum_{x \in C} |\text{ccl}(x) \cap K|$. Since $K \trianglelefteq G$, each conjugacy class is either disjoint from K or contained in K , so $p \mid \sum |\text{ccl}(x) \cap K|$. As $p \mid |K|$, we get $p \mid |Z(G) \cap K|$, and the same argument as (i) gives $Z(G) \cap K$ non-trivial.

(iii) Let M be a maximal normal subgroup of G contained in H . Then G/M is a p -group, so by (i) it has a non-trivial centre $L/M \trianglelefteq G/M$. By the Correspondence Theorem, $L \trianglelefteq G$. Since $M \subsetneq L$ strictly, maximality of M forces $L \not\subseteq H$.

It remains to show $L \subseteq N_G(H)$. To show $\ell \in N_G(H)$ we need $\ell^{-1}h\ell \in H$ for all $h \in H$. Take $h \in H$ and $\ell \in L$. Since $\ell M \in L/M = Z(G/M)$, it commutes with every coset, so $(\ell M)(hM) = (hM)(\ell M)$, i.e. $\ell hM = h\ell M$. This gives $\ell^{-1}h\ell \in hM$. Since $h \in H$ and $M \subseteq H$, we have $hM \subseteq H$, so $\ell^{-1}h\ell \in H$. Hence $\ell \in N_G(H)$, so $L \subseteq N_G(H)$.

Since $L \subseteq N_G(H)$ but $L \not\subseteq H$, we have $H \subsetneq N_G(H)$. Finally, if $|H| = p^{n-1}$ then $N_G(H)$ has order strictly greater than p^{n-1} , and the only possibility is $p^n = |G|$, so $N_G(H) = G$ and $H \trianglelefteq G$. $\square$

Theorem 13 (2.2 — Burnside's Lemma). *Let G be a finite group acting on a finite set X , and for $g \in G$ let $X_g := \{x \in X \mid gx = x\}$. Then the number k of orbits in X under G is*

$$k = \frac{1}{|G|} \sum_{g \in G} |X_g|.$$

Proof. Let $S := \{(g, x) \in G \times X \mid gx = x\}$. Fixing g gives $|S| = \sum_{x \in X} |X_g|$; fixing x gives $|S| = \sum_{g \in G} |G_x|$. Let $C \subseteq X$ contain one element from each orbit. By the Orbit-Stabiliser Theorem, $|G_x| = |G|/|Gx| = |G|/|Ga|$ for all $x \in Ga$. Hence

$$\frac{1}{|G|} \sum_{g \in G} |X_g| = \frac{|S|}{|G|} = \frac{1}{|G|} \sum_{x \in X} |G_x| = \frac{1}{|G|} \sum_{a \in C} \sum_{x \in Ga} \frac{|G|}{|Ga|} = \frac{1}{|G|} \sum_{a \in C} |G| = k. \quad \square$$

Remark 1 (Pólya counting). *Burnside's Lemma counts distinct objects under symmetry. As an illustration: how many necklaces with p beads (p prime) can be made from n colours, where two necklaces are the same if one is a rotation of the other?*

Let X be the set of all colourings, so $|X| = n^p$. The cyclic group $G = \langle a \rangle$ of order p acts on X by rotation. For the identity, $X_e = X$ so $|X_e| = n^p$. For any $g \neq e$: since $|G|$ is prime, g generates G , so a necklace fixed by g must be fixed by all rotations, meaning all beads have the same colour. Hence $|X_g| = n$. Burnside then gives

$$k = \frac{1}{p} (n^p + (p-1)n).$$

Lecture 3

Definition 2 (p -group, Sylow p -subgroup). *Let p be a prime. A group G is a p -group if every element of G has order a power of p . If G is finite and $p^m \mid |G|$ but $p^{m+1} \nmid |G|$, then any subgroup of G of order p^m is called a Sylow p -subgroup of G .*

Theorem 14 (3.5 — First Sylow Theorem). *Let G be a finite group and p a prime. If $p^m \mid |G|$, then G has a subgroup of order p^m .*

Proof. If $m = 0$ the result is trivial. We proceed by induction on $n = |G|$, the case $n = 1$ being trivial. Assume the result holds for all groups of order less than n .

Case 1: $p \mid |Z(G)|$. By the Fundamental Theorem of Finitely Generated Abelian Groups, $Z(G)$ contains an element a of order p . Let $C = \langle a \rangle$. Since $a \in Z(G)$, we have $C \trianglelefteq G$, and $|G/C| = n/p$, which is divisible by p^{m-1} . By the induction hypothesis, G/C has a subgroup of order p^{m-1} , which by the Correspondence Theorem is of the form H/C for some $H \leq G$. Then $|H| = |H/C| \cdot |C| = p^{m-1} \cdot p = p^m$.

Case 2: $p \nmid |Z(G)|$. From the class equation $n = |Z(G)| + \sum_a |G : C_G(a)|$, since $p \mid n$ but $p \nmid |Z(G)|$, there must exist some $a \notin Z(G)$ with $p \nmid |G : C_G(a)|$. Since $|G| = |G : C_G(a)| \cdot |C_G(a)|$ and $p^m \mid |G|$ but $p \nmid |G : C_G(a)|$, it follows that $p^m \mid |C_G(a)|$. Since $a \notin Z(G)$, we have $C_G(a) \neq G$, so $|C_G(a)| < n$. By the induction hypothesis $C_G(a)$ contains a subgroup of order p^m , which is also a subgroup of G . $\square$

Corollary 2 (3.6 — Cauchy's Theorem). *If p is a prime dividing $|G|$, then G has an element of order p .*

Proof. By the First Sylow Theorem, G has a subgroup H of order p . Every non-identity element of H has order p . $\square$

Corollary 3 (3.7). *A finite group G is a p -group if and only if $|G|$ is a power of p .*

Proof. If $|G|$ is a power of p , then by Lagrange every element order divides $|G|$, hence is a power of p . Conversely, if $|G|$ is not a power of p , then some prime $q \neq p$ divides $|G|$, and by Cauchy's Theorem G has an element of order q , contradicting that G is a p -group. $\square$

Theorem 15 (3.8 — Second Sylow Theorem). *Let G be a finite group and p a prime. All Sylow p -subgroups of G are conjugate.*

Proof. Let $S, H \in \text{Syl}_p(G)$. Let S act on the coset space G/H by left translation: $s \cdot gH = sgH$.

Orbits have p -power size. By the Orbit-Stabiliser Theorem, each orbit has size $|S : S_{gH}|$, which divides $|S| = p^m$, hence is a power of p .

There must be a fixed point. Since H is a Sylow p -subgroup, $p \nmid |G|/|H| = |G/H|$. Writing $|G/H|$ as a sum of orbit sizes, each of which is a power of p , the only way this sum is not divisible by p is if at least one orbit has size $p^0 = 1$.

A fixed point gives conjugacy. If gH is a fixed point, then $s \cdot gH = gH$ for all $s \in S$, so $g^{-1}sg \in H$ for all $s \in S$. Hence $g^{-1}Sg \subseteq H$, and since $|g^{-1}Sg| = |S| = |H|$, we get $g^{-1}Sg = H$. So S and H are conjugate. $\square$

Corollary 4 (3.10). *A Sylow p -subgroup of G is unique if and only if it is normal in G .*

Proof. If P is the unique Sylow p -subgroup, then $gPg^{-1} = P$ for all $g \in G$ (since conjugates are also Sylow p -subgroups), so $P \trianglelefteq G$. Conversely, if $P \trianglelefteq G$ then P is its own conjugate, so by the Second Sylow Theorem it is the only Sylow p -subgroup. $\square$

Theorem 16 (3.9 — Third Sylow Theorem). *Let G be a finite group and p a prime. The number n_p of Sylow p -subgroups satisfies $n_p \mid |G|$ and $n_p \equiv 1 \pmod{p}$.*

Proof. $n_p \mid |G|$. By the Second Sylow Theorem, all Sylow p -subgroups are conjugate to a fixed $K \in \text{Syl}_p(G)$, so $n_p = |\text{ccl}(K)| = |G : N_G(K)|$, which divides $|G|$.

$n_p \equiv 1 \pmod{p}$. Let $P \in \text{Syl}_p(G)$. Since conjugates of Sylow p -subgroups are again Sylow p -subgroups (by the Second Sylow Theorem), P acts on the set $\text{Syl}_p(G)$ by conjugation:

$$x \cdot Q := xQx^{-1}, \quad x \in P, \quad Q \in \text{Syl}_p(G).$$

This action partitions $\text{Syl}_p(G)$ into orbits. Let $F \subseteq \text{Syl}_p(G)$ denote the set of fixed points (those Q with $xQx^{-1} = Q$ for all $x \in P$), and let $\mathcal{O}$ denote the collection of orbits of size greater than 1. Then:

$$n_p = |\text{Syl}_p(G)| = |F| + \sum_{O \in \mathcal{O}} |O|.$$

By the Orbit-Stabiliser Theorem each $|O|$ divides $|P| = p^m$, so every $O \in \mathcal{O}$ has size p^e for some $e \geq 1$, hence $p \mid |O|$. Therefore $n_p \equiv |F| \pmod{p}$.

It remains to show $F = \{P\}$, which gives $n_p \equiv 1 \pmod{p}$. Clearly $P \in F$. If $Q \in F$, then $xQx^{-1} = Q$ for all $x \in P$, so $P \leq N_G(Q)$. By Corollary 3.10, since $Q \trianglelefteq N_G(Q)$, Q is the unique Sylow p -subgroup of $N_G(Q)$. But P is also a Sylow p -subgroup of $N_G(Q)$, so $P = Q$. Hence $F = \{P\}$. $\square$